90° Pipe Brace Bracket: Static FEA Report

Part: 90degree_12in_0.133_Pipe_Brace_Bracket_V1 (Default<As Machined>) | Study: Static 1 | Date: November 3, 2026

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1. Summary

This bracket is a simple metal support: a 1 in NPS Schedule 40 steel pipe, bent 90° on a 12 in centerline radius, welded between two 4 × 4 × 3/8 in bolted plates. Whatever mounts to the end plate gets its load carried back into the floor, wall or frame the base is bolted to. The goal of this study was to find out how the bracket handles a 1,000 N (225 lbf) load and where it is weakest.

Result	Value	Comment
Peak von Mises stress	375.9 MPa (54.5 ksi)	Localized at pipe-to-plate junction; exceeds yield
Nominal bend stress (FEA field)	≈110–190 MPa	Lower bend near base plate
Nominal bend stress (hand calc)	140 MPa (20.3 ksi)	M = F·R at base, S = 0.1328 in³
Max resultant displacement	4.26 mm (0.168 in)	At free end plate
Max equivalent strain	1.09×10^{-3}	Element 4311
Factor of safety (yield)	0.59 peak / 1.2–2.0 nominal	Yield 220.6 MPa
Total mass	2.54 kg (5.61 lb)	Two bodies, plain carbon steel

2. Geometry

Feature	Dimension
Pipe	1 in NPS Sch 40: OD 1.315 in, ID 1.049 in, wall 0.133 in
Bend	90°, 12.0 in centerline radius (≈18.8 in arc length)
Mounting plates (2)	4.00 × 4.00 × 0.375 in, 0.10 in corner radius
Bolt holes	Ø0.531 in (17/32 in) thru, 4 per plate (8 total), 2.50 in square pattern on plate center; clearance for 1/2 in bolts
Pipe bore in end plate	Ø1.330 in thru, centered (0.0075 in radial clearance on 1.315 in OD)
Bodies	Body 1: pipe + base plate, 1.891 kg, 2.424×10^{-4} m³. Body 2: end plate, 0.653 kg, 8.374×10^{-5} m³



Figure 1. Model with fixtures (green, base plate) and load (magenta, end plate).

3. Setup

Material: Plain Carbon Steel (linear elastic isotropic)

E = 210 GPa, v = 0.28, G = 79 GPa, ρ = 7,800 kg/m³, Sy = 220.6 MPa (32.0 ksi), Su = 399.8 MPa (58.0 ksi), α = 1.3 × 10^{-5} /K. Real pipe and plate would likely be ASTM A53 Gr. B (Sy ≈ 240 MPa) and A36 (Sy ≈ 250 MPa), so using this material is slightly conservative.

Loads and fixtures

Item	Entity	Type	Reaction (N)
Fixed-1	4 bolt-hole bores, base plate	Fixed geometry	0.71
Roller/Slider-1	Underside of base plate	Roller/slider	997.3
Force-1	Face of end plate	1,000 N along ref. edge (vertical)	n/a

The reactions add up to 999.999 N against the 1,000 N applied, with a free-body residual of 0.0021 N, so the model is in equilibrium. Almost all of the load goes through the base plate bearing on the surface below it, which is what you would expect for a bolted base.

Interaction and mesh

Global contact: bonded, independent mesh (end plate bonded to pipe). Blended curvature-based solid mesh, high quality (16 Jacobian points): 41,720 nodes, 23,845 elements, element size 0.077–0.589 in. 93.5% of elements have aspect ratio < 3, 0.36% exceed 10 (max 430), 0% distorted. Mesh time 4 s. Large displacement, friction, soft spring and inertial relief off; thermal effects on at 298 K zero-strain reference with no temperature load applied.

4. Assumptions

1. Linear elastic, small-displacement, static response; no plasticity, buckling, fatigue or dynamic amplification.
2. Material is homogeneous, isotropic plain carbon steel at room temperature (298 K); no thermal gradient, so thermal stress is zero.
3. The welds are idealized. The pipe and base plate are one solid body and the end plate is fully bonded, so there are no weld fillets, heat-affected zone or residual stresses in the model.
4. The bolts are not modeled. Fixing the hole bores and adding a roller on the bearing face stands in for tightened 1/2 in bolts on a rigid surface. A real mounting surface and bolt stretch would make the bracket somewhat more flexible.
5. The 1,000 N load is uniformly distributed over the end-plate face and acts vertically. Self-weight (24.9 N) is neglected. Results scale linearly with load and apply equally to a downward load of the same magnitude.
6. Nominal pipe dimensions; no corrosion allowance, mill tolerance or bend-induced wall thinning or ovality.

5. Results

Stress builds smoothly along the bend and is highest near the base, where the moment arm is longest (about 12 in). The 375.9 MPa maximum (node 18098) is a tiny hot spot right at the sharp corner where the pipe meets the plate. Sharp corners like this produce artificially high stress in FEA, and the number keeps climbing as the mesh gets finer, so it is not the real stress level in the pipe. What it does show is that the weld at the base is the most likely place for the bracket to fail.

Hand check: To make sure the FEA was reasonable, I treated the bend as a curved cantilever. M = 1,000 N × 0.3048 m = 304.8 N·m and S = π(1.315⁴ − 1.049⁴)/(32 × 1.315) = 0.1328 in³ = 2.18 × 10^{-6} m³, so σ = M/S = 140 MPa, which falls inside the 110–190 MPa range the FEA shows at the bend. For deflection, a quarter-circle cantilever gives δ = πFR³/(4EI) = 2.9 mm with I = 0.0873 in⁴. The FEA gives 4.26 mm, which makes sense because my hand calc treats the base as perfectly rigid, while in the model the base plate can flex between the bolts and the end plate can rotate.

Bolts: if the base plate tips about its edge, each of the two tension-side bolts carries about 1.85 kN (415 lbf). A 1/2-13 Grade 5 bolt is good for about 53 kN, so the bolts are not a concern.



Figure 2. von Mises stress (deformation scale 11.1×). Peak 3.759 × 10⁸ N/m² at the pipe-plate junction.

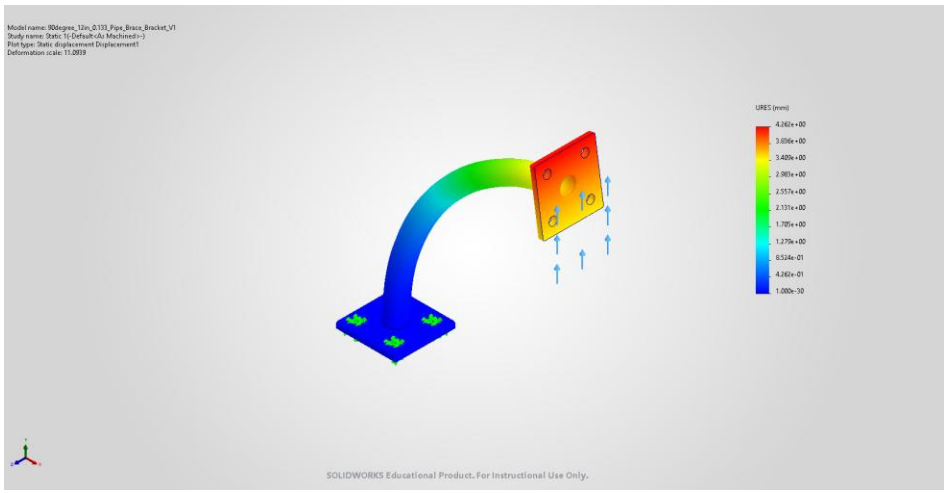


Figure 3. Resultant displacement. Maximum 4.262 mm at the free end plate.

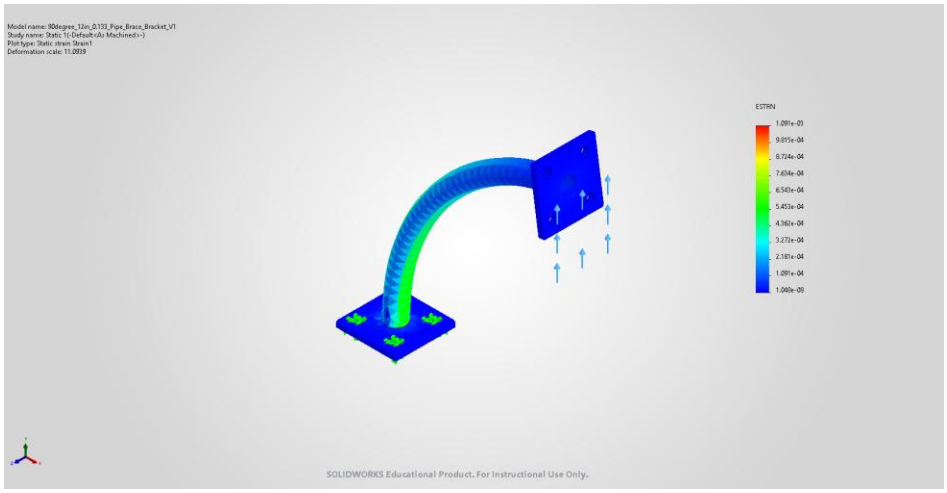


Figure 4. Equivalent strain. Maximum 1.091 × 10 ⁻³.

6. Conclusions

7. **The pipe holds 1,000 N, but without much margin.** The 140 MPa bending stress gives a factor of safety of about 1.6 on yield, or about 1.2 using the high end of the FEA range. That is a little under the 1.67 commonly used for steel in allowable-stress design.

8. **The weld at the base is the weak spot.** The 376 MPa peak (FOS 0.59) is exaggerated by the sharp corner, but it still suggests the metal right at the weld could start to yield. If the load repeated, from vibration for example, this is where a fatigue crack would most likely start.
9. **Deflection is fine for a brace.** 4.3 mm (0.17 in) over a roughly 12 in reach is about L/72. That is acceptable for a static support, but too much if whatever it holds needs to stay precisely aligned.
10. **The plates and bolts are not the problem.** The 3/8 in plates and 1/2 in bolts have plenty of margin. The pipe bend controls the design.
11. **What I would change.** I would keep this version under about 700 N (155 lbf) to stay above a safety factor of 1.67. To keep the full 1,000 N, switching to 1 in Sch 80 pipe ($\sigma \approx 116$ MPa, FOS 1.9) or 1-1/4 in Sch 40 ($\sigma \approx 79$ MPa, FOS 2.8) would solve it. Adding a full fillet weld (at least 3/16 in) or a gusset at the base would help too. As a next step, I would model the weld fillet and refine the mesh at that corner to get a better read on the real peak stress.